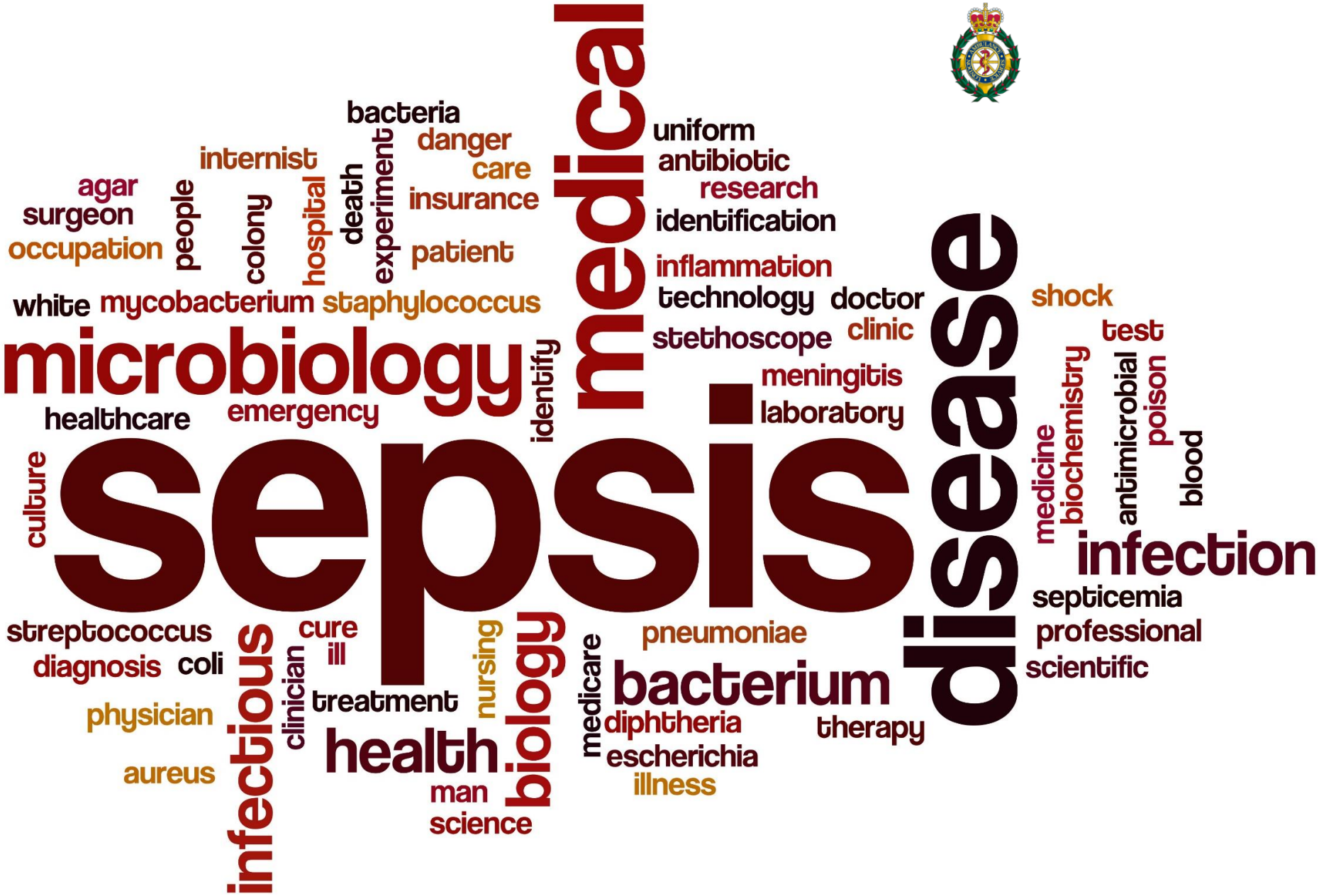
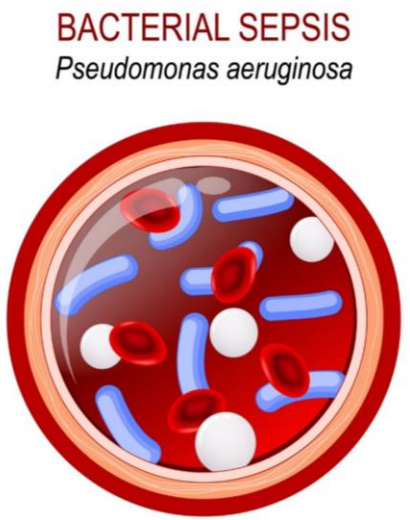
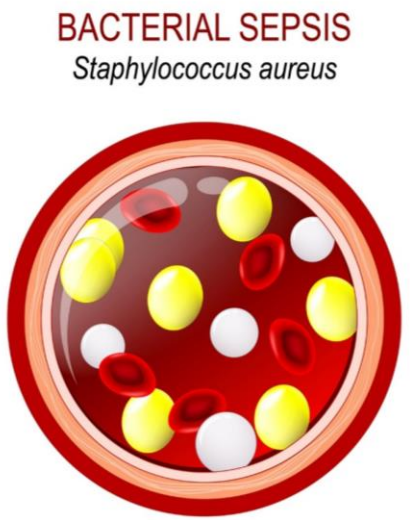

sepsis
medical
disease
infection
microbiology
bacterium
health
biology
infectious
bacteria
danger
internist
care
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surgeon
people
colony
hospital
death
experiment
insurance
occupation
patient
uniform
antibiotic
research
identification
inflammation
technology
doctor
shock
white
mycobacterium
staphylococcus
stethoscope
clinic
test
identify
meningitis
laboratory
healthcare
emergency
culture
medicine
biochemistry
antimicrobial
poison
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septicemia
professional
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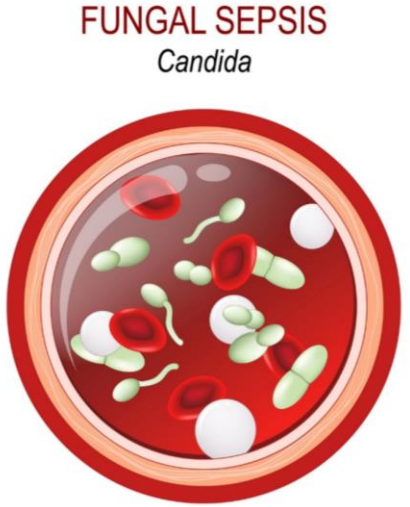
Sepsis/ Suspected sepsis

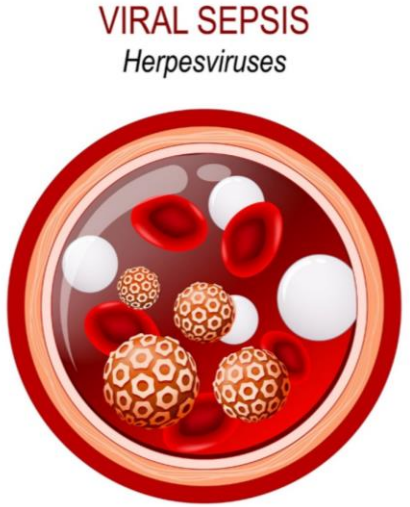
- **Sepsis**
- Sepsis is a life-threatening organ dysfunction
- This is due to a dysregulated host response to infection





- **Suspected sepsis**
- Suspected Sepsis is not a specific illness
- It is a syndrome





Introduction

- Sepsis is a clinical syndrome caused by the body's immune and coagulation systems being switched on by an infection.

- Sepsis with shock is a life-threatening condition, characterised by
  - Low blood pressure despite adequate fluid replacement
  - Organ dysfunction or failure

Immune response

- Sometimes, the immune system's response goes out of control, causing widespread inflammation throughout the body.

- Inflammation is meant to help fight the infection
- Too much inflammation can harm the body's own tissues and organs

Frequency

- **5 deaths per hour in the UK**
- **48,000** people **died** from sepsis last year (**2023-24**)
- **40%** of people who develop sepsis are estimated to suffer physical, cognitive and/ or psychological after effects following recovery
- Long term complications due to delayed diagnosis were found to be costing the UK economy as much as £15.6 billion annually

Frequency

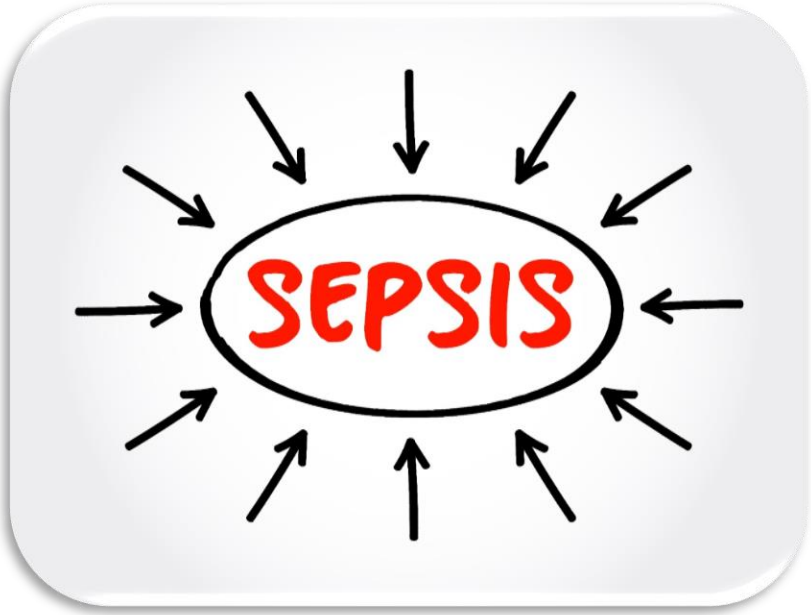
- **Mortality rate is high**
  - (36% for severe sepsis and up to 50% in septic shock)
- You are six times more likely to die from sepsis than ACS or stroke
- Sepsis **can arise** from **any infection**
- It claims 11 million lives globally every year (more than bowel, breast, and prostate cancer combined)

Blood vessel tone/ integrity

- Infection causes blood vessels to lose ability to constrict/ dilate to regulate BP
- Inflammation in sepsis can damage lining of blood vessels, making them ‘leaky’
- This causes the fluid shift into tissues
- Swelling, tachycardia, SoB, hypotension can result

Organ Failure

- Low blood pressure and leaky vessels reduce the amount of blood and oxygen reaching the organs
- Organs start to fail because they are not getting the oxygen and nutrients they need to function
- Multiple organ failure is often what leads to death in severe cases of sepsis


SEPSIS

Disseminated Intravascular Coagulation

- The body's inflammation response can cause **abnormal clotting** in the blood vessels

- Also known as **Disseminated Intravascular Coagulation** or **DIC**

- ***2 stages:***

1. Over active clotting can block blood flow, causing organ damage
2. Platelets are used up, meaning the body cannot stop the bleeding process

Abnormal Clotting

- This can lead to excessive, catastrophic bleeding from infection or even minor injury

- Examination *may* reveal:
- Petechiae, hypoxia, hypotension, GI bleeding, confusion

- Early recognition/ suspicion and rapid transfer is critical

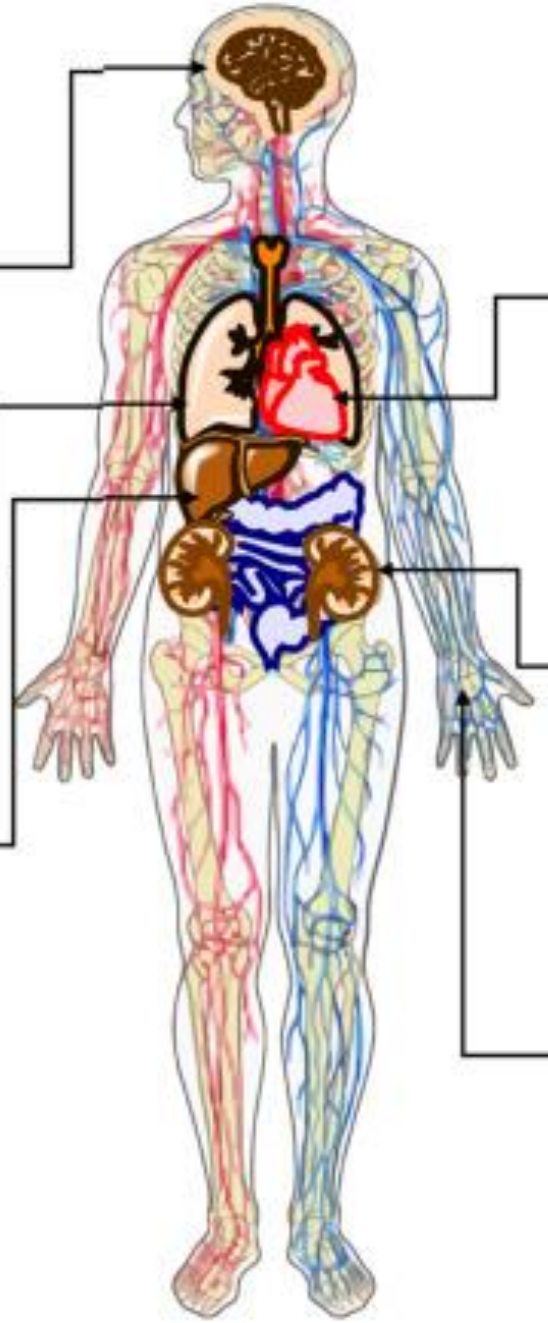




Central nervous system
- Altered consciousness
- Confusion
- Psychosis

Respiratory system
- Hypoxia
- Tachypnea
- ↓ PaO_2 :FiO_2 ratio

Hepatic system
- Jaundice
- ↑ Bilirubin levels
- ↑ Liver enzymes
- ↓ Albumin



Cardiovascular system
- Tachycardia
- Hypotension
- ↑ Lactate levels
- Altered echocardiography variables

Renal system
- Oliguria
- ↑ Creatinine
- ↑ Blood urea nitrogen

Hematological system
- ↓ Platelets
- Petechiae
- ↓ Protein C
- ↑ D-dimer
- Disseminated intravascular coagulation

Some causes of sepsis

- Meningitis/ encephalitis
- Pneumonia/ likely chest source
- Urinary tract infection, abdominal pain or distension
- Cellulitis, septic arthritis, infected wound
- Device related infection (e.g. urinary catheter, IV device)
- Infection can be suspected but source not obvious

Groups at Risk

- Over 75
- Impaired immune systems
- Chemotherapy treatments
- DAMN drugs (**D**iuretics, **A**CE inhibitors, **M**etformin, **N**SAIDs)
- Surgery in past 6 weeks
- Indwelling lines/ catheters
- Breach of skin (burns, blisters, skin infection)
- IV drug use

Groups at Risk

- Women who have just given birth or recently been pregnant
  - (Including those who have had a miscarriage or abortion)

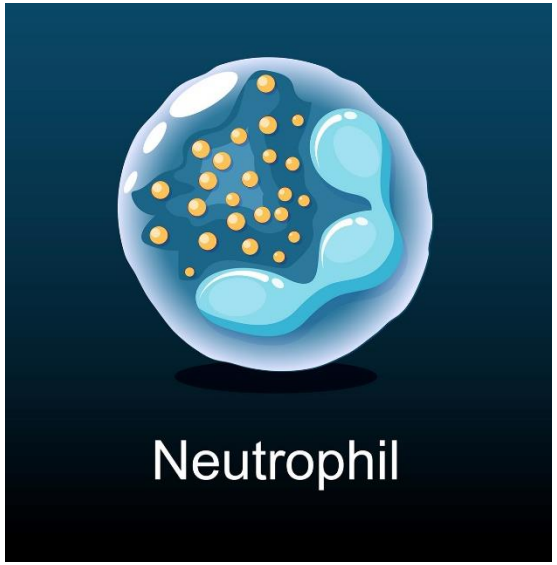
- **NEWS2 is not to be used with pregnant patients**

- Babies younger than 1 year
- Babies are also more at risk if they were born prematurely or their mother had an infection while she was pregnant.
  - This can be any type of infection, including very mild ones.

Neutrophils

Neutrophils are a type of white blood cell

They play a key role in fighting infection



Some medical treatments can suppress the production of WBCs, leading to a reduced ability to fight infection.

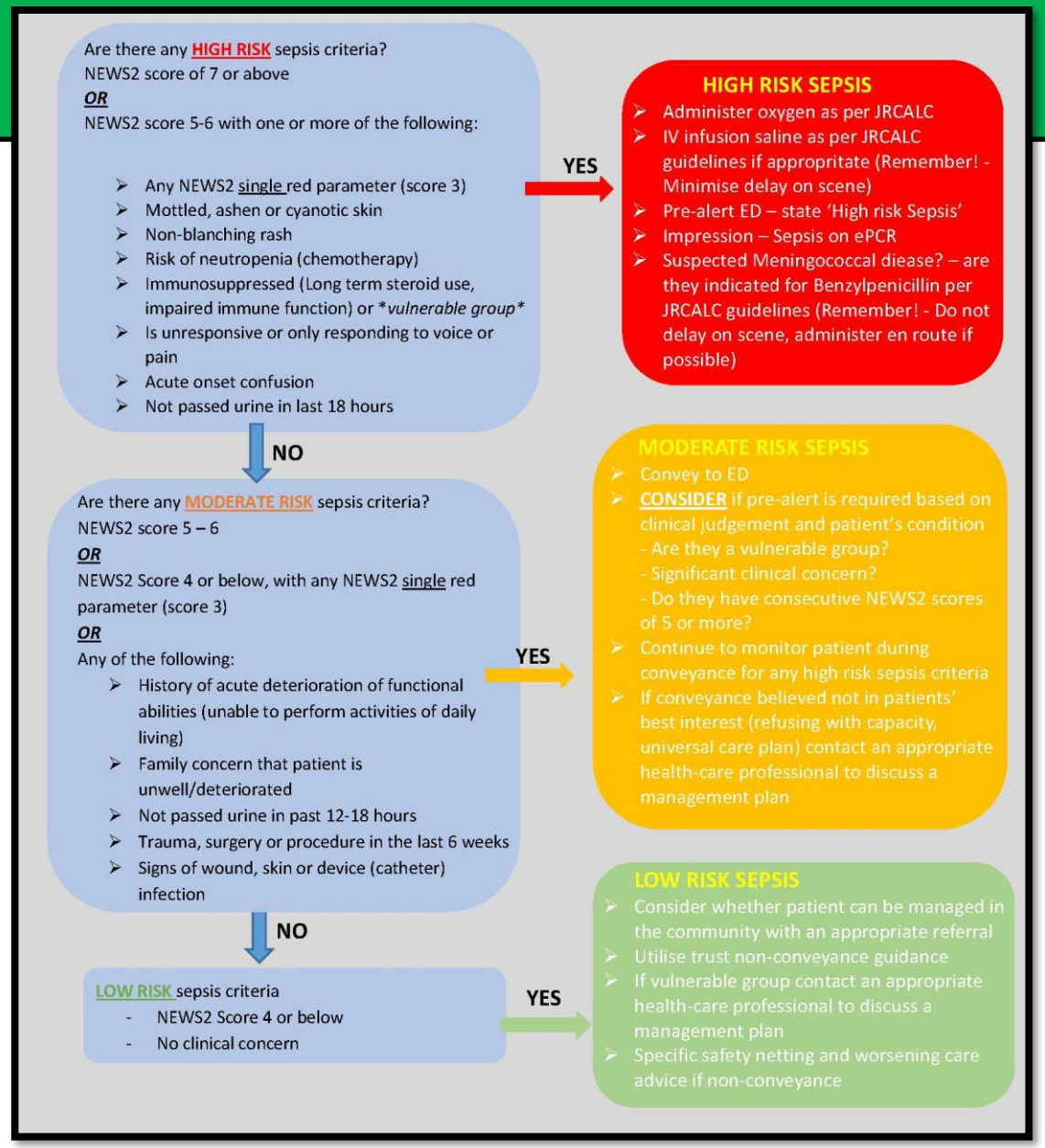
Neutropaenic sepsis

- A reaction to an infection that affects the whole body
- A potentially life-threatening complication of anticancer and other immunosuppressive drug treatment
- Defined as a temperature of greater than 38°C or any symptoms and/ or signs of sepsis with a lowered neutrophil count

Neutropaenic sepsis

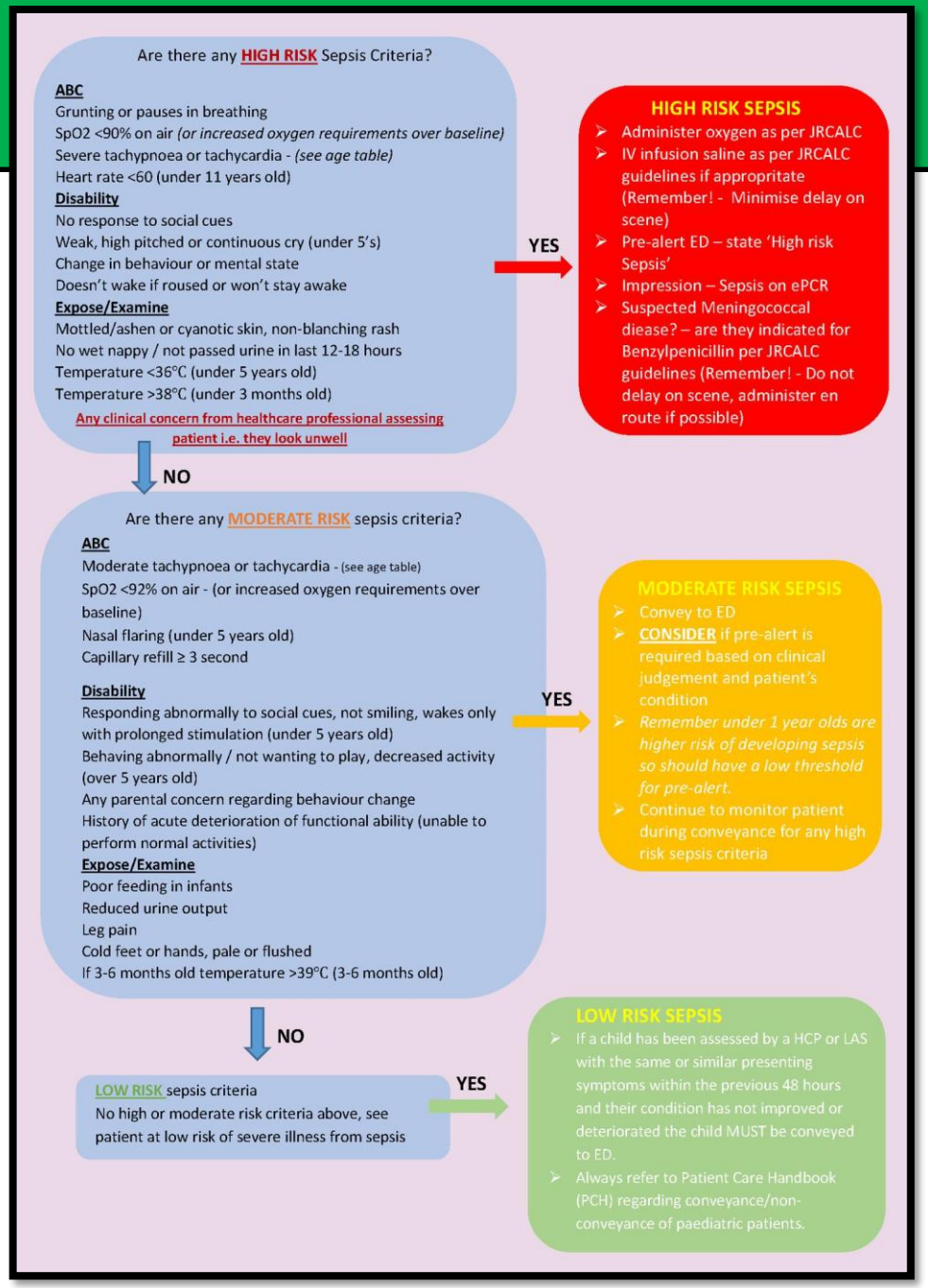
- Neutropaenic patients are at higher risk because their body's ability to fight infections is compromised
- When they get an infection, their body struggles to fight it off, potentially leading to sepsis
- The same dangerous cycle of inflammation, blood clots, leaky vessels, and organ damage occurs, but can progress faster because the immune system is weakened

Adult (16 years and over) Sepsis screening tool



Adults:
Use in conjunction with a NEWS2 score

PAED (16 years and under)
Sepsis screening tool



NEWS2

Physiological parameter	Score 3	2	1	0	1	2	3
Respiration rate (per minute)	≤8		9–11	12–20		21–24	≥25
SpO_2 Scale 1 (%)	≤91	92–93	94–95	≥96			
SpO_2 Scale 2 (%)	≤83	84–85	86–87	88–92 ≥93 on air	93–94 on oxygen	95–96 on oxygen	≥97 on oxygen
Air or oxygen?		Oxygen		Air			
Systolic blood pressure (mmHg)	≤90	91–100	101–110	111–219			≥220
Pulse (per minute)	≤40		41–50	51–90	91–110	111–130	≥131
Consciousness				Alert			CVPU
Temperature (°C)	≤35.0		35.1–36.0	36.1–38.0	38.1–39.0	≥39.1	

Primary survey

- **D:**
- **R:** Any confusion or reduced LoC?
- **A:** Cyanosis?
- **B:** Increased rate? Dib? Infection related?
- **C:** Tachycardia? Irregular? CRT prolonged?
- **D:** PEARL? Fast? MSCx4? Temp (limited value)
- **E:** Evaluate- could this sepsis?

Key areas: Secondary survey

- **History**:
  - SAMPLE
  - Fever/ infection? Duration?
  - PMHx (Ca, immune problems, recent injury)
  - SHx, DHx
  - SOCRATES
  - Any triggers?
- **Examination**
  - Ausc chest
  - Rashes?
  - RoS
- **Observations**
  - LoC: monitor for confusion
  - Pallor/ hypoxia/ cyanosis
  - RR: Is it elevated?
  - HR: Tachy or irregular?
  - BP: unreliable indicator of sepsis
  - Use Sepsis tool
- Oxygen? NOO?
- Senior clinician
- (Benzyl-Pen, IV PAR, NaCl)

Suspect sepsis in:

- Pt with fever/ unwell **and**
- NEWS2 greater than or equal to 5 **and/ or**
- looks unwell with a history of infection.

- NEWS2 does not diagnose sepsis

- Keep on-scene times to a minimum.

- Provide a pre-alert and NEWS2 score if ?sepsis

Thank you. Do you have any questions?

Suspect SEPSIS Save Lives

THE UK SEPSIS TRUST

7th edition
Created 2024